

ESERCIZIO 1

$$\begin{aligned} \min \quad & 2x_1 + 3x_2 - 4x_3 \\ \text{s.t.} \quad & 3x_1 - x_2 + x_3 + x_4 = 2 \\ & 4x_1 + 2x_2 - 2x_3 - 4x_4 = 4 \\ & 5x_1 - 2x_3 - 3x_4 = -1 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

FORMA STANDARD

$$\begin{aligned} \min \quad & 2x_1 + 3x_2 - 4x_3 \\ \text{s.t.} \quad & 3x_1 - x_2 + x_3 + x_4 - x_5 = 2 \\ & 4x_1 + 2x_2 - 2x_3 - 4x_4 + x_6 = 4 \\ & 5x_1 - 2x_3 - 3x_4 = -1 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{aligned}$$

MATRICE COEFFICIENTI

$$A = \begin{pmatrix} 3 & -1 & 1 & 1 & -1 & 0 \\ 4 & 2 & -2 & -4 & 0 & 1 \\ 5 & 0 & -2 & -3 & 0 & 0 \end{pmatrix}$$

VARIABILI ARTIFICIALI

$$\begin{aligned} 3x_1 - x_2 + x_3 + x_4 - x_5 + y_4 &= 2 \\ 4x_1 + 2x_2 - 2x_3 - 4x_4 + x_6 &= 4 \\ 5x_1 - 2x_3 - 3x_4 + y_8 &= -1 \\ x_1, x_2, x_3, x_4, x_5, x_6, y_4, y_8 &\geq 0 \end{aligned}$$

MATRICE COEFFICIENTI

$$A = \begin{pmatrix} 3 & -1 & 1 & 1 & -1 & 0 & 1 & 0 \\ 4 & 2 & -2 & -4 & 0 & 1 & 0 & 0 \\ 5 & 0 & -2 & -3 & 0 & 0 & 0 & 1 \end{pmatrix}$$

x_6, y_4, y_8
matrice identità

PROBLEMA ARTIFICIALE

$$\begin{aligned} \min \quad & y_4 + y_8 \\ \text{s.t.} \quad & 3x_1 - x_2 + x_3 + x_4 - x_5 + y_4 = 2 \\ & 4x_1 + 2x_2 - 2x_3 - 4x_4 + x_6 = 4 \\ & 5x_1 - 2x_3 - 3x_4 + y_8 = -1 \\ & x_1, x_2, x_3, x_4, x_5, x_6, y_4, y_8 \geq 0 \end{aligned}$$

TABEAU

	x_1	x_2	x_3	x_4	x_5	x_6	y_4	y_8	$\bar{b}$
2	0	0	0	0	0	0	1	1	0
x_6	3	-1	1	1	-1	0	1	0	2
y_4	4	2	-2	-4	0	1	0	0	4
y_8	5	0	-2	-3	0	0	0	1	1

$$-3 + \left(\frac{4}{5} \cdot 3 - \frac{2}{5} \cdot 3 \right)$$

$$-3 + \left(\frac{21}{5} - \frac{15}{5} \right)$$

$$-3 + \frac{6}{5}$$

$$-\frac{15}{5} + \frac{6}{5} = -\frac{9}{5}$$

$$\begin{array}{cccc} 0 & -6 & -6 & 0 \\ -3 & 3 & 0 & -3 \end{array}$$

$$3 + (0-3) = 0$$

$$4 + (-6+3) = 1$$

$$6 + (-6+0) = 0$$

$$0 + (0-3) = -3$$

$$\begin{array}{cccc} 4 & -3 & -6 & 0 \\ 0 & 4/5 & 1 & 0 \\ 1 & -2/5 & 0 & 1 \end{array}$$

$$\cdot 6 = 0 \begin{array}{cccc} 4/5 & 6 & 0 \end{array}$$

$$\cdot -4 = -4 \begin{array}{cccc} 16/5 & 0 & -4 \end{array}$$

$$4 + (0-4) = 0$$

$$-3 + \left(\frac{6}{5} + \frac{6}{5} \right) = \frac{6}{5}$$

$$-6 + (6+0) = 0$$

$$0 + (0-4) = -4$$

$$\Rightarrow \begin{array}{cccc} 0 & 4/5 & 0 & -4 \\ 0 & 4/5 & 1 & 0 \\ 1 & -2/5 & 0 & 1 \end{array}$$